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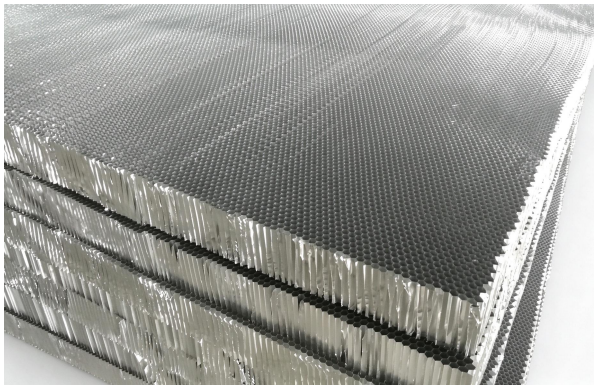
A2 R1 BS8414-1✔ RoHS

3000 Series aluminum honeycomb core

Aluminum Honeycomb Core

Aluminum honeycomb core is adhesively—bonded of the aluminum foil,which have sharp and clear surface. We produce micro-aperture honeycomb core and regular size core.

It is a lightweight, high-strength sandwich core material with advantages of low density, high stiffness, flatness, fire resistance and corrosion resistance. It is widely used for composite sandwich panels across industries:



Expanded honeycomb

Features

- High strength-to-weight ratio
- High surface area-to-volume ratio
- Lightweight and high strength
- Moisture resistant
- Corrosion resistant
- Fire resistant
- Green material



Unepxanded slices

Available in Various Categories:

Block

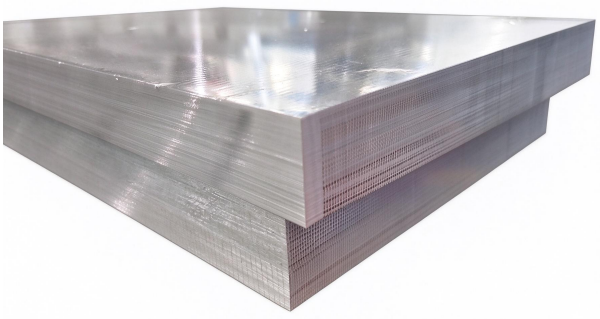
Unexpanded slices

Expanded sheets

Perforated or Non-perforated

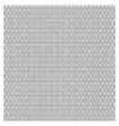
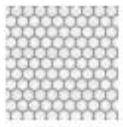
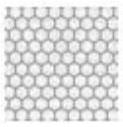
Drilled core

Mill finish or corrosion treated



Blocks

Specifiation options

Micro-aperture cellsize							
Cell sizes avallable							
	1/30"	1/25"	1/15"	1/10"	1/8"	1/6"	1/5"
	0.87mm	1.04mm	1.73mm	2.6mm	3.17mm	3.47mm	5.2mm
Alu foil thk(mm)	0.025,0.03	0.025,0.03	0.04	0.04,0.05,0.06		0.04,0.05,0.06 0.07,0.08,0.1	
Reguar cell size							
Cell sizes avallable							
	1/4"	1/3"	3/8"	1/2"	3/4"	4/5"	1"
	6.35mm	8.47mm	9.53mm	12.7mm	19.05mm	20.32mm	25.9mm
Alu foil thk(mm)	0.04,0.05,0.06,0.07,0.08,0.1,0.13						

Additional densities and cell sizes available upon request.

Node strenggh

Density kg/m³	Temperature 23±5°C		After boiling for 2 hours		Temperature 150±5°C		Temperature 175±5°C	
	Average value kN/m	Min value kN/m	Average value kN/m	Min value kN/m	Average value kN/m	Min value kN/m	Average value kN/m	Min value kN/m
15-166	1.47	1.32	1.32	1.25	0.64	0.6	0.55	0.5

Compressive strength and shear strength

Cellsize	**Foil thickness mm**	**Density kg/m³**	**Temperature 23±5°C**		
			Compressive strength Mpa	**Shear strength Mpa "L"**	**Shear strength Mpa "W"**
1/8"	0.03	66	2.39	1.70	1.00
1/8"	0.04	77	3.10	2.18	1.25
1/5"	0.03	44	1.18	0.89	0.52
1/5"	0.04	52	1.60	1.15	0.67

1/5"	0.05	68	2.50	1.78	1.04
1/5"	0.06	81	3.35	2.37	1.33
1/5"	0.08	110	5.47	3.68	2.23
1/5"	0.10	140	8.53	4.99	2.98
1/4"	0.03	33	0.73	0.58	0.33
1/4"	0.04	39	0.98	0.75	0.43
1/4"	0.05	53	1.65	1.18	0.69
1/4"	0.06	61	2.07	1.48	0.86
1/4"	0.08	84	3.57	2.50	1.42
1/4"	0.10	106	5.29	3.51	2.12
1/3"	0.03	27	0.53	0.44	0.24
1/3"	0.04	31	0.66	0.53	0.30
1/3"	0.05	41	1.07	0.80	0.47
1/3"	0.06	49	1.43	1.03	0.60
1/3"	0.08	67	2.45	1.74	1.02
1/3"	0.10	85	3.67	2.54	1.49
3/8"	0.06	42.03	1.18	0.89	0.52
3/8"	0.07	49.04	1.43	1.03	0.60
1/2"	0.06	33.63	0.73	0.58	0.33
1/2"	0.07	39.23	0.98	0.75	0.43

The above data is based on various sample sizes and is for reference only.

Quality standard

Material	Aluminum honeyomb
Alloy	3003H18
Standard	China Aviation Industry Standard HB5443-90
Temperature range applied	(-55～250)°C in long term
Cell size tolerance	within ±5%
Density tolerance	within ±5%
Complaint with	ROHS,Reach,DIN 5510-2:2009 & DIN 54837:2007